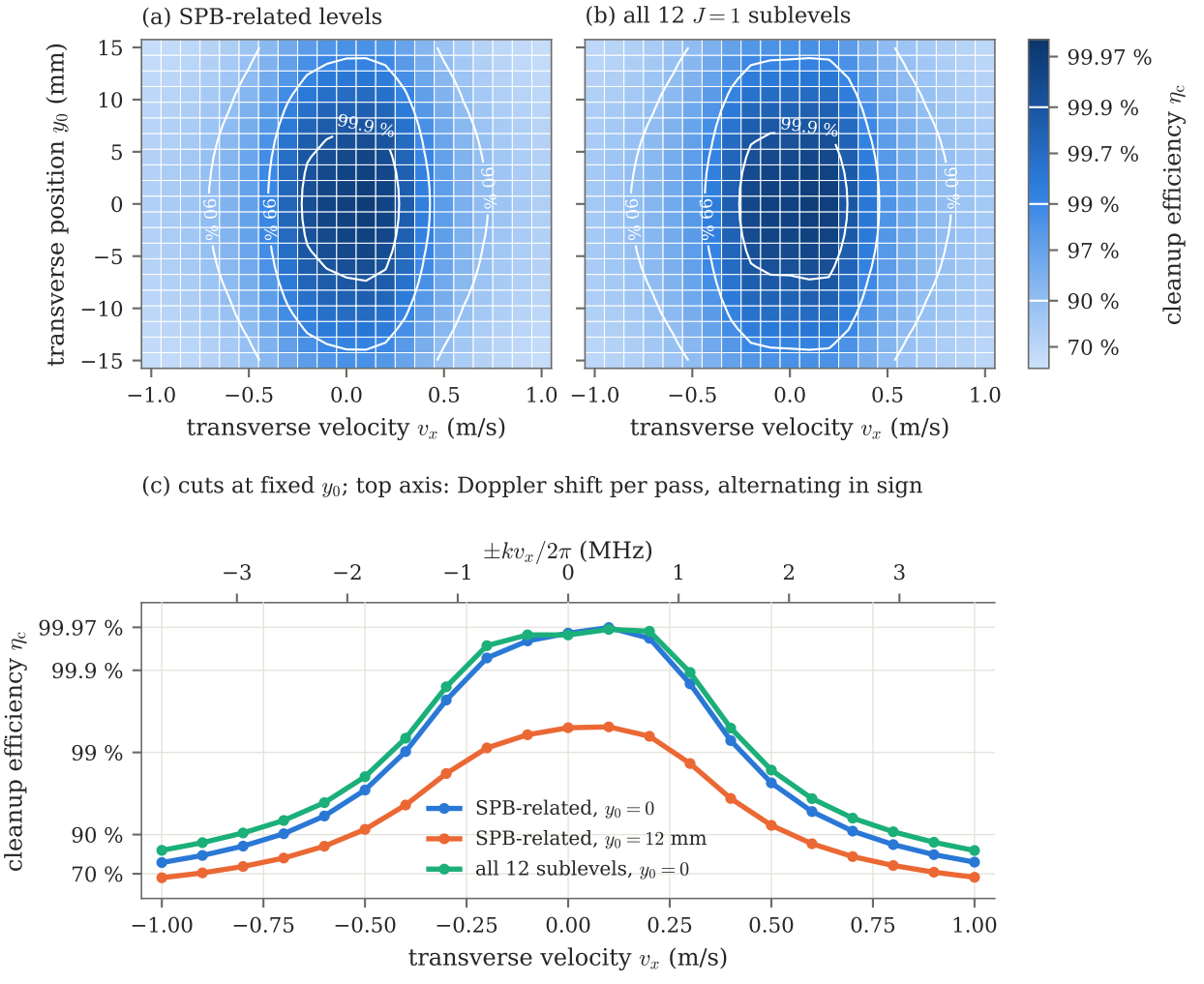

(a) SPB-related levels
(b) all 12 $J=1$ sublevels
transverse position y_0 (mm)
transverse velocity v_x (m/s)
cleanup efficiency η_c
99.97 %
99.9 %
99.7 %
99 %
97 %
90 %
70 %
(c) cuts at fixed y_0; top axis: Doppler shift per pass, alternating in sign
$\pm k v_x/2\pi$ (MHz)
SPB-related, $y_0 = 0$
SPB-related, $y_0 = 12$ mm
all 12 sublevels, $y_0 = 0$